

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

A quadratic function models the height, in feet, of an object above the ground in terms of the time, in seconds, after the object is launched off an elevated surface. The model indicates the object has an initial height of **10** feet above the ground and reaches its maximum height of **1,034** feet above the ground **8** seconds after being launched. Based on the model, what is the height, in feet, of the object above the ground **10** seconds after being launched?

Answer

- A. **234**
- B. **778**
- C. **970**
- D. **1,014**

Correct Answer: C

Rationale

Choice C is correct. It's given that a quadratic function models the height, in feet, of an object above the ground in terms of the time, in seconds, after the object is launched off an elevated surface. This quadratic function can be defined by an equation of the form $f(x) = a(x - h)^2 + k$, where $f(x)$ is the height of the object x seconds after it was launched, and a , h , and k are constants such that the function reaches its maximum value, k , when $x = h$. Since the model indicates the object reaches its maximum height of **1,034** feet above the ground **8** seconds after being launched, $f(x)$ reaches its maximum value, **1,034**, when $x = 8$. Therefore, $k = 1,034$ and $h = 8$. Substituting **8** for h and **1,034** for k in the function $f(x) = a(x - h)^2 + k$ yields $f(x) = a(x - 8)^2 + 1,034$. Since the model indicates the object has an initial height of **10** feet above the ground, the value of $f(x)$ is **10** when $x = 0$. Substituting **0** for x and **10** for $f(x)$ in the equation $f(x) = a(x - 8)^2 + 1,034$ yields $10 = a(0 - 8)^2 + 1,034$, or $10 = 64a + 1,034$. Subtracting **1,034** from both sides of this equation yields $64a = -1,024$. Dividing both sides of this equation by **64** yields $a = -16$. Therefore, the model can be represented by the equation $f(x) = -16(x - 8)^2 + 1,034$. Substituting **10** for x in this equation yields $f(10) = -16(10 - 8)^2 + 1,034$, or $f(10) = 970$. Therefore, based on the model, **10** seconds after being launched, the height of the object above the ground is **970** feet.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.